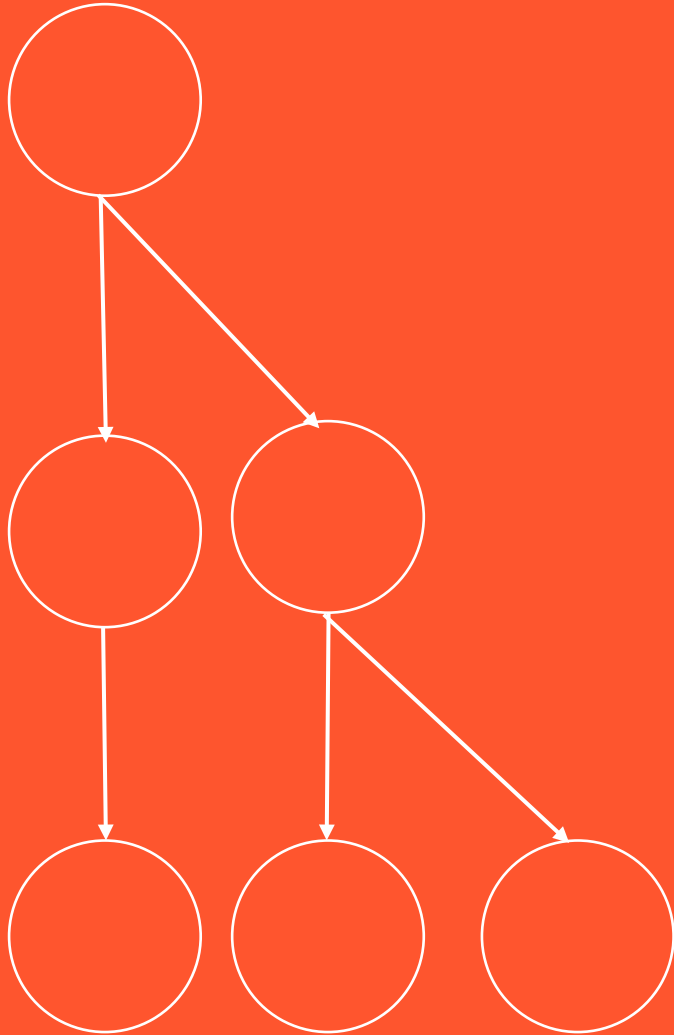


BCOG 100

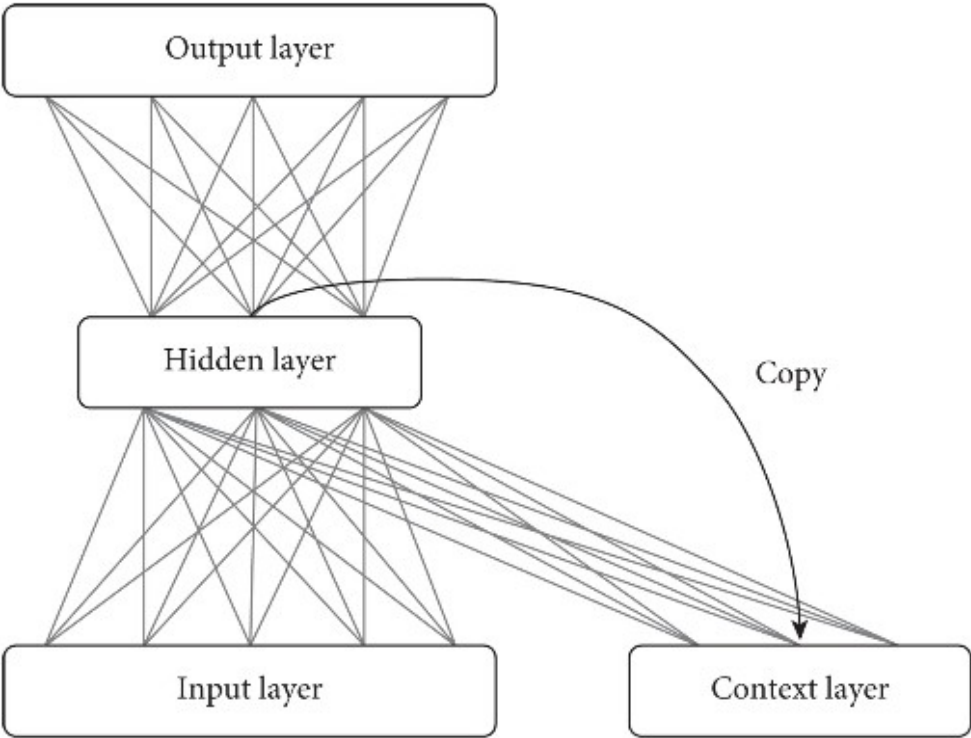
Pattern Recognition

Lecture 3: Brain Implementation

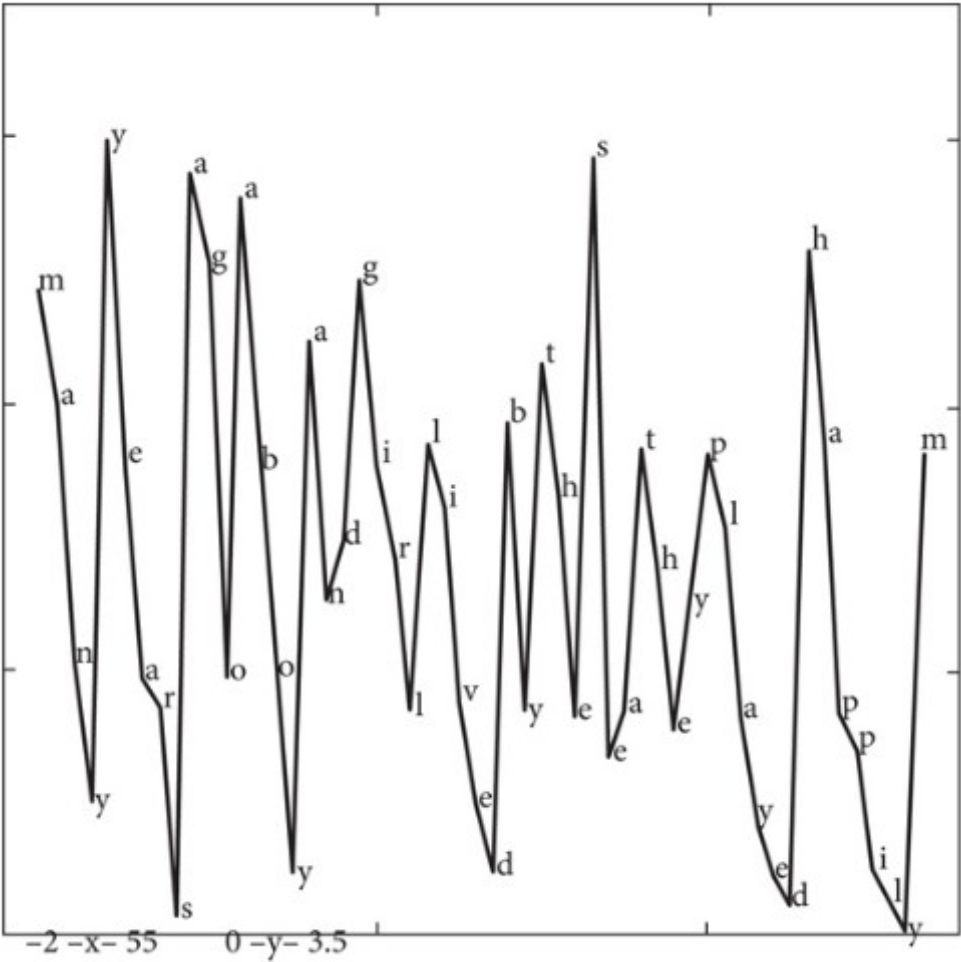


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Model Architecture



Click to view larger



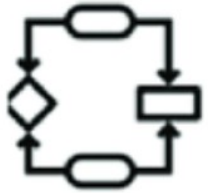
MANYYEARSAGOABOYANDGIRLLIVEDBYTH
ESEA

Pattern Recognition in the Brain



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

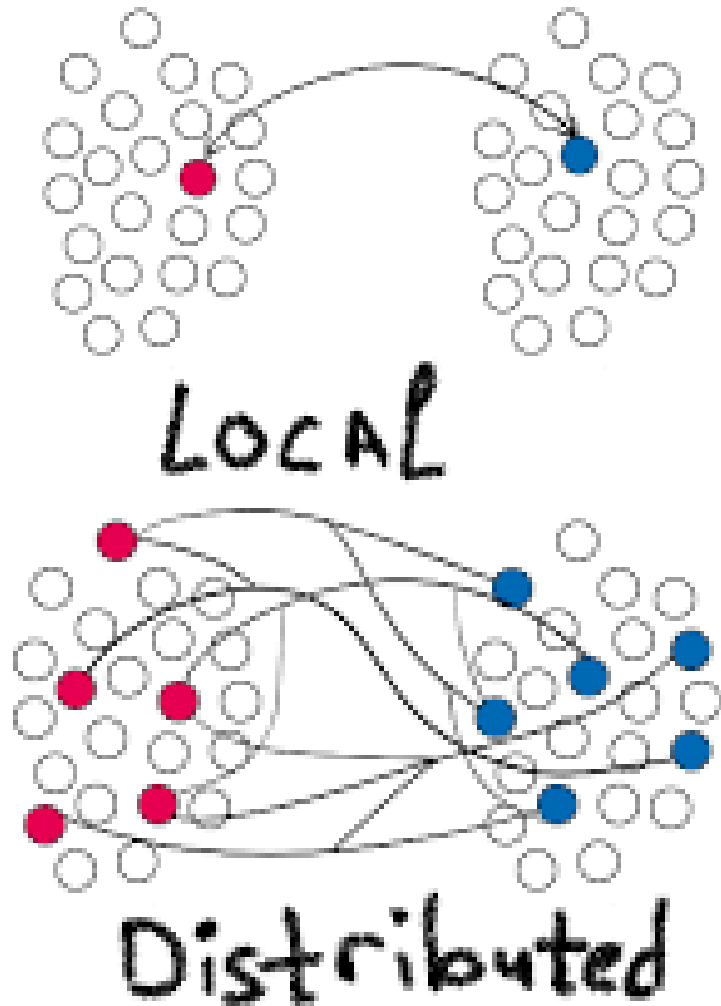
Recap & Overview

The problems of pattern recognition, and algorithms for dealing with those problems

Today:

- How brain systems implement those algorithms
- Neural mechanisms, sensory systems, and brain structures

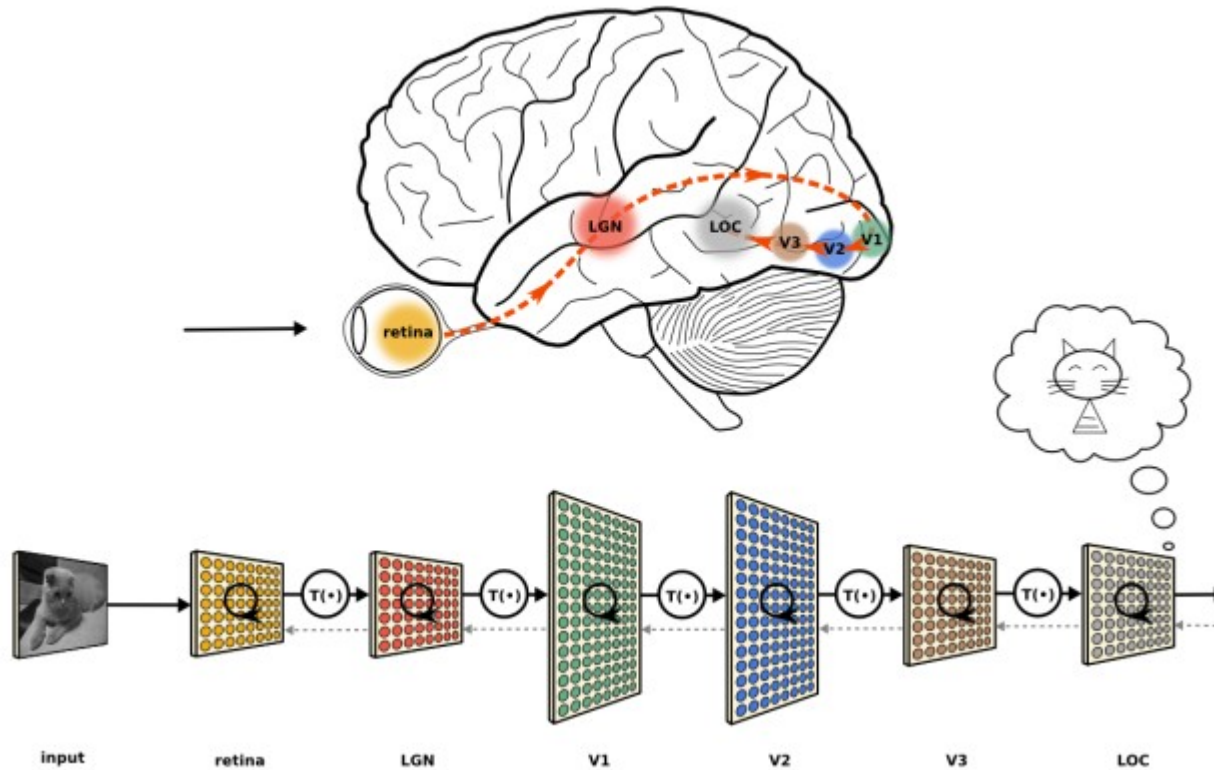
Pattern Recognition in the Brain



Distributed and Sparse Coding

- Patterns across populations
- Sparse activation
- Efficiency and robustness

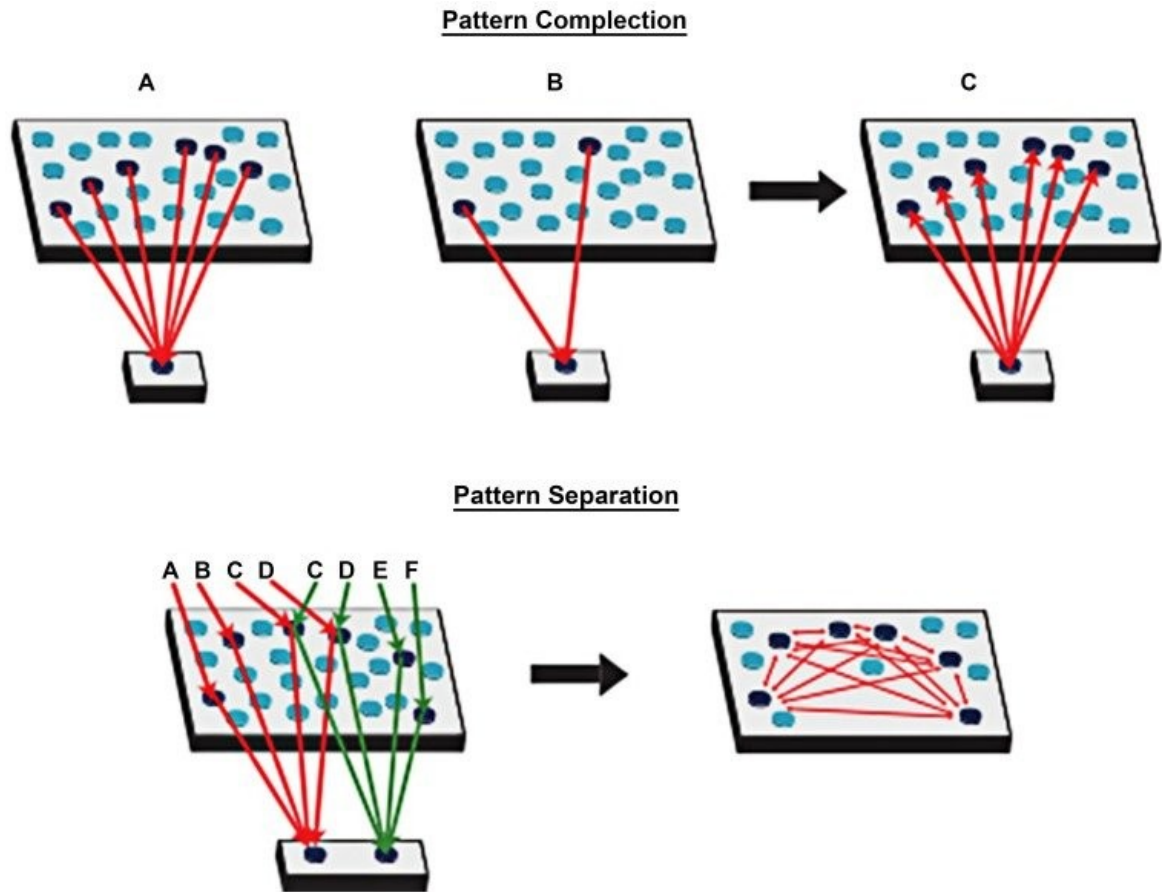
Pattern Recognition in the Brain



Hierarchical Representation and Abstraction

- Simple to complex
- Feature combinations
- Abstraction and invariance

Pattern Recognition in the Brain



Pattern Separation and Completion

- Keeping memories distinct
- Filling in missing input
- Hippocampal mechanisms

Pattern Recognition in the Brain

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MicroNetwork Motifs

A. Feedforward excitation



B. Feedforward inhibition



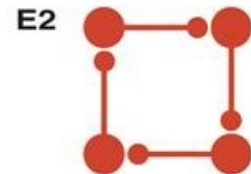
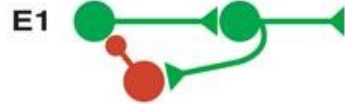
C. Convergence/divergence



D. Lateral inhibition



E. Feedback/Recurrent inhibition



F. Feedback/Recurrent excitation



—▶ Excitation
—● Inhibition

Recurrent and Predictive Dynamics

- Feedback loops
- Prediction and error
- Learning over time

Figure 6

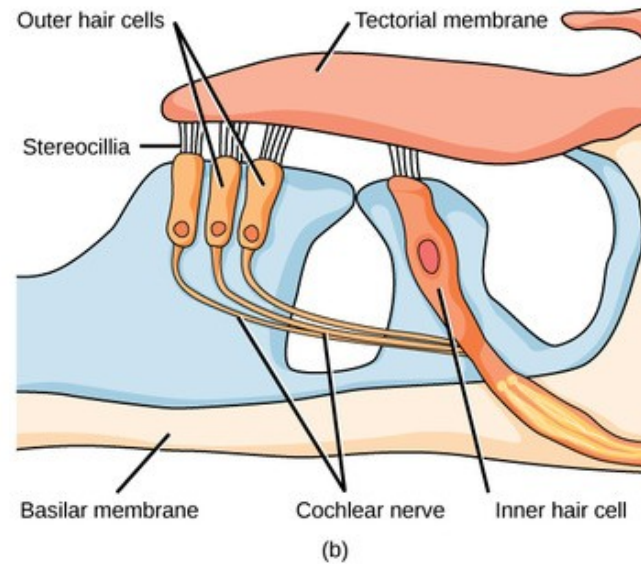
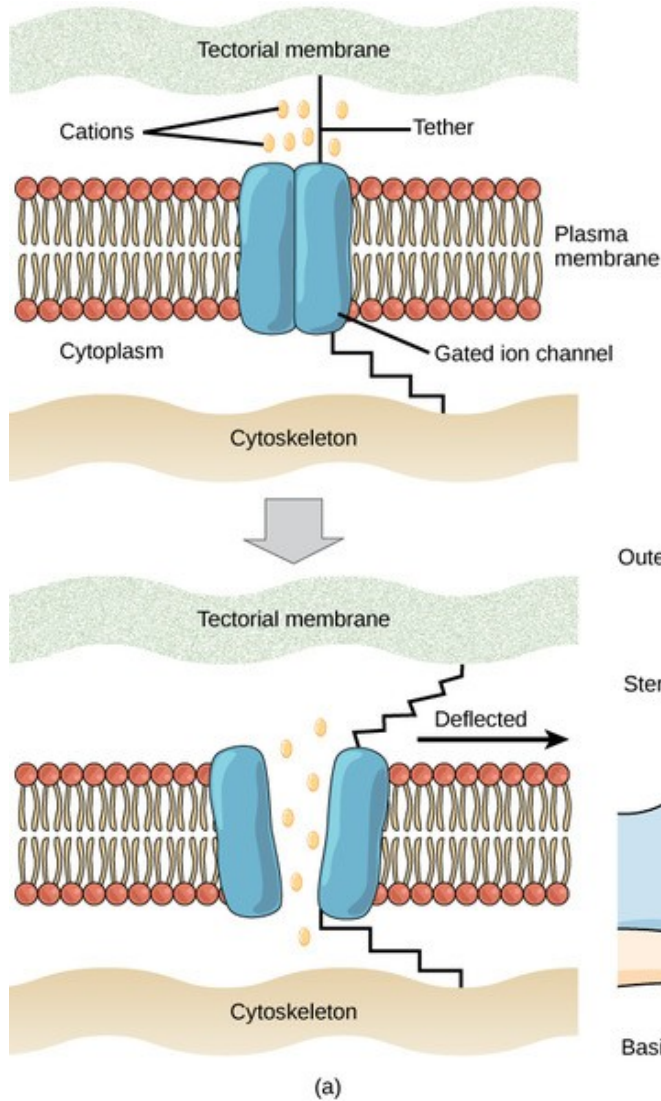
Pattern Recognition in the Brain



Sensory Evolution: From Detection to Pattern Recognition

- Ancient origins
- Specialized organs
- Topographic organization

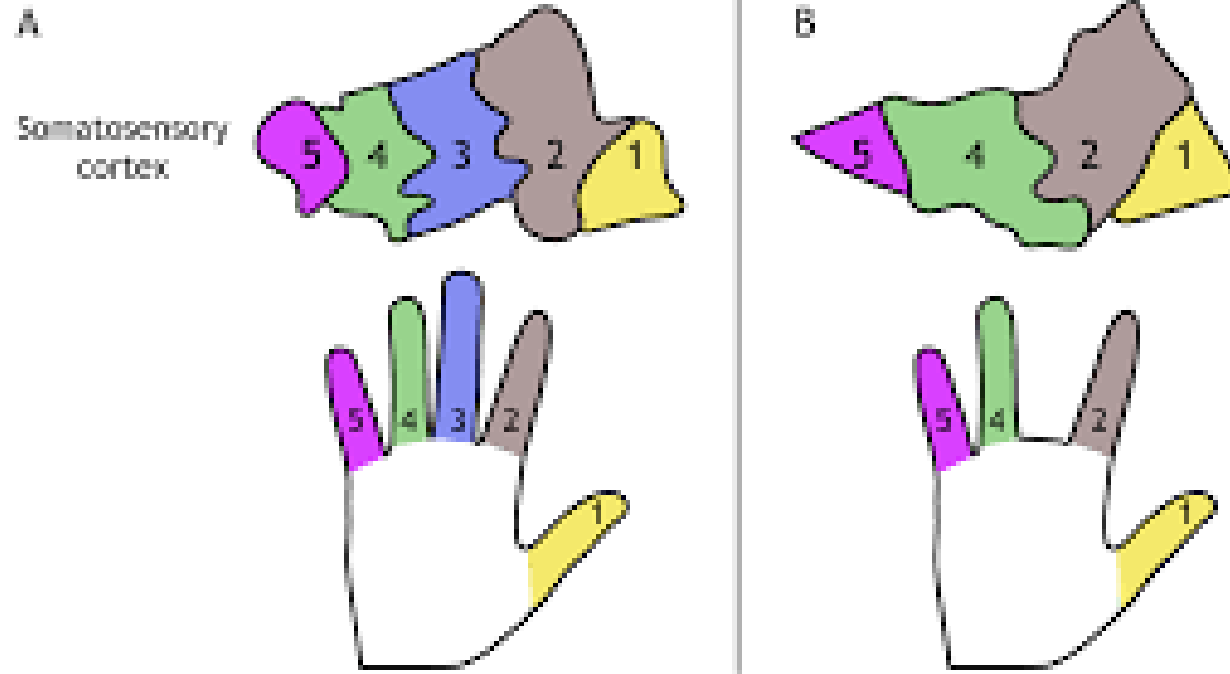
Pattern Recognition in the Brain



Somatosensory Transduction

- Physical deformation → electrical signal
- Ion channels and receptor diversity
- From surface contact to neural code

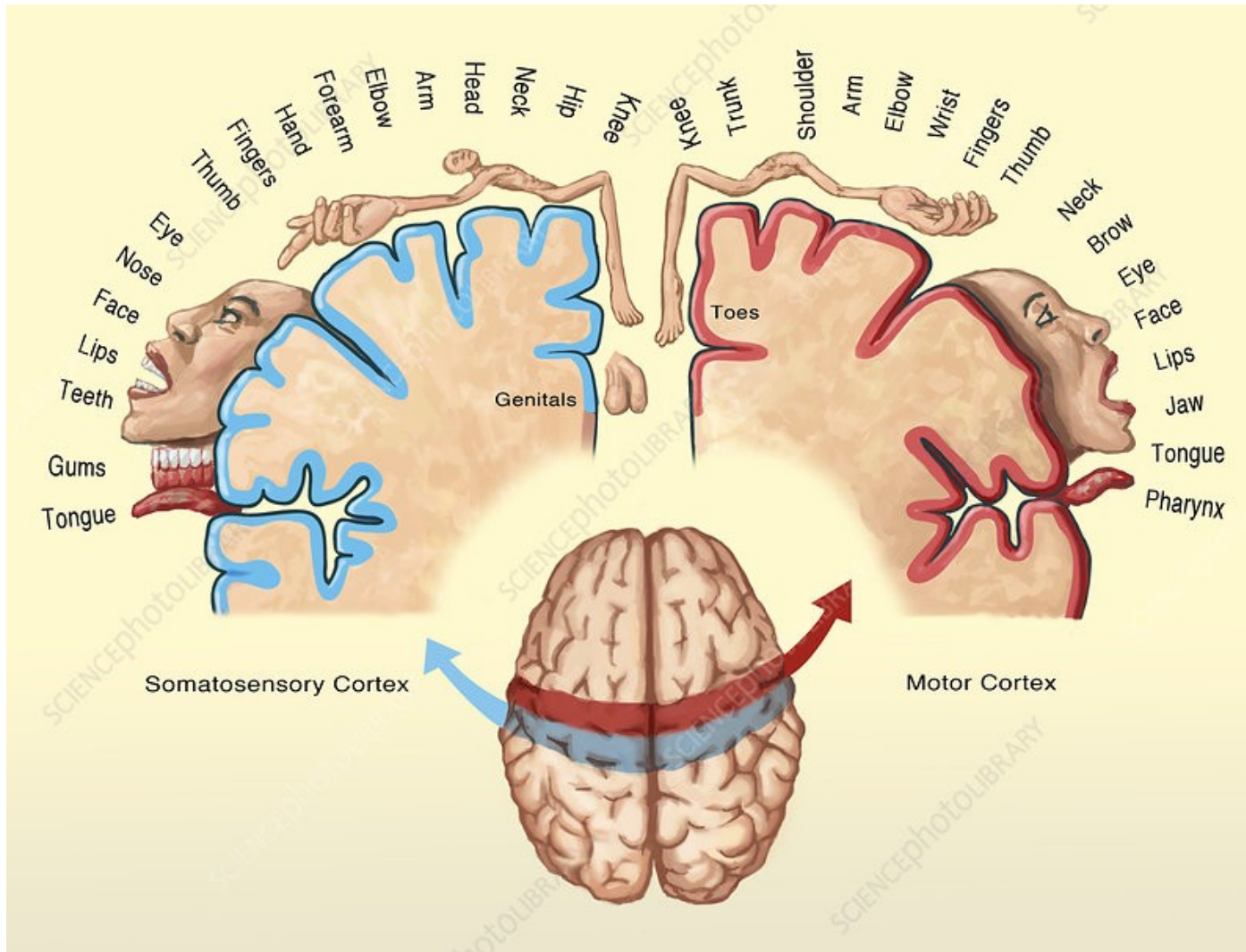
Pattern Recognition in the Brain



Somatosensation Pattern Formation

- Spatial mapping
- Body geometry preserved
- Local feature detection

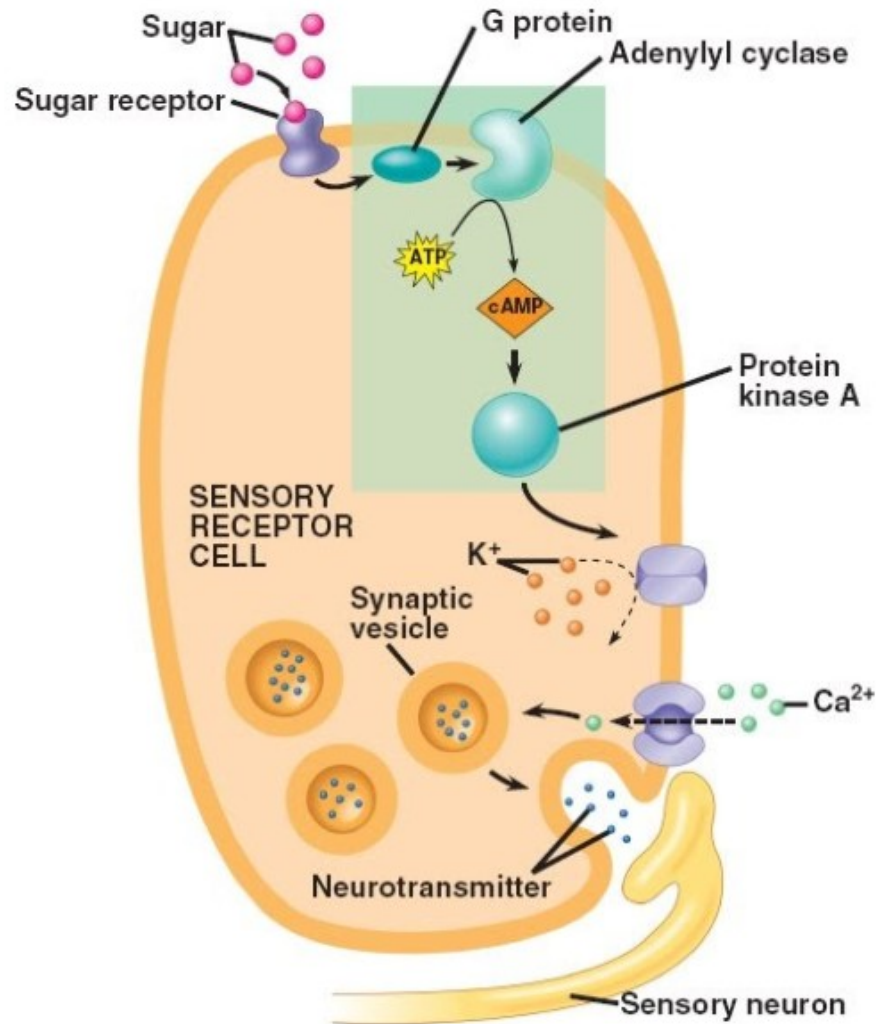
Pattern Recognition in the Brain



Somatosensation in the Brain

- Spinal and brainstem circuits
- Tectum and pallium
- Precursors of sensory neocortex

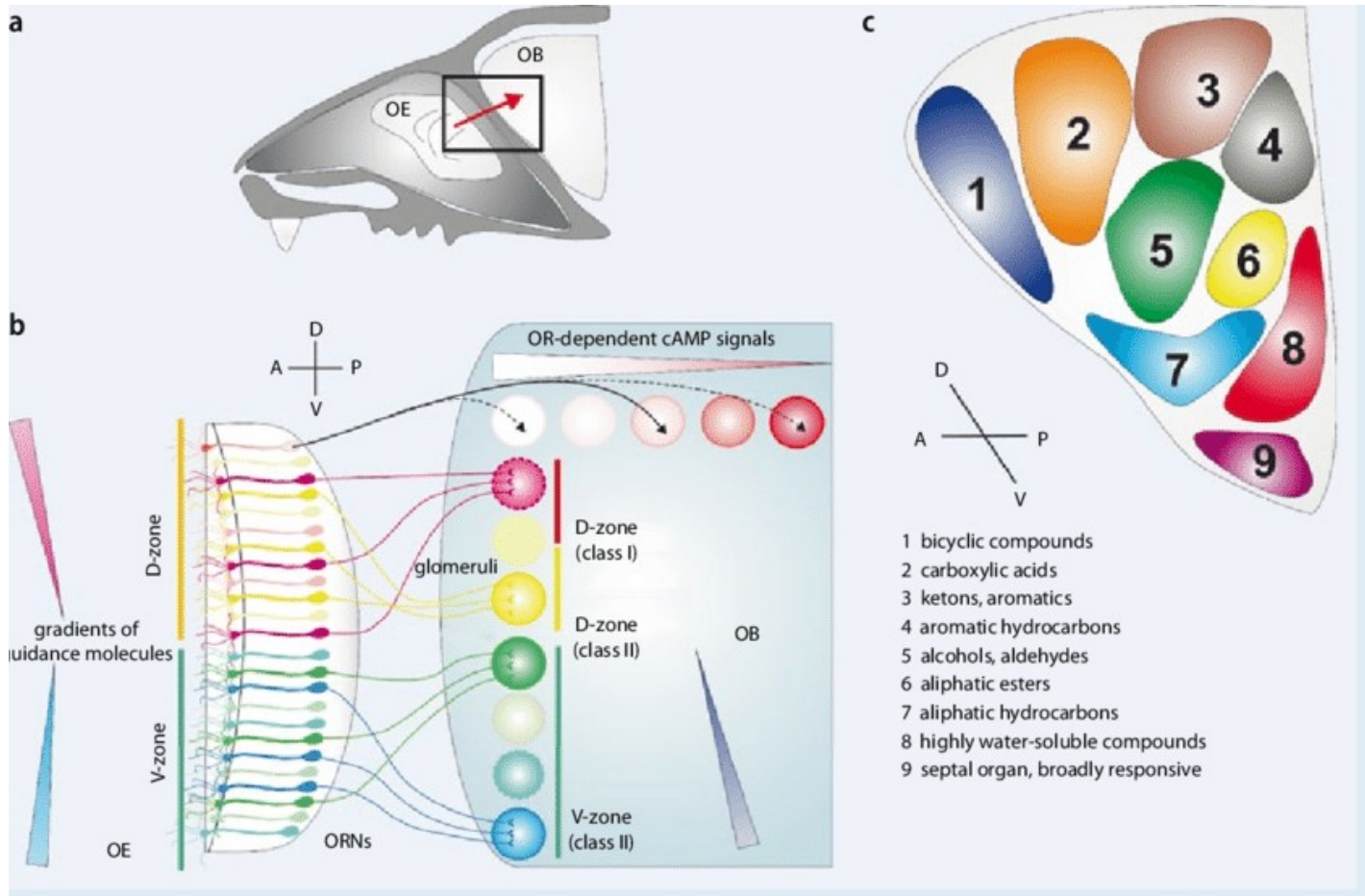
Pattern Recognition in the Brain



Chemosensory Transduction for Smell and Taste

- Molecular binding
- Receptor diversity
- Translating chemistry to electricity

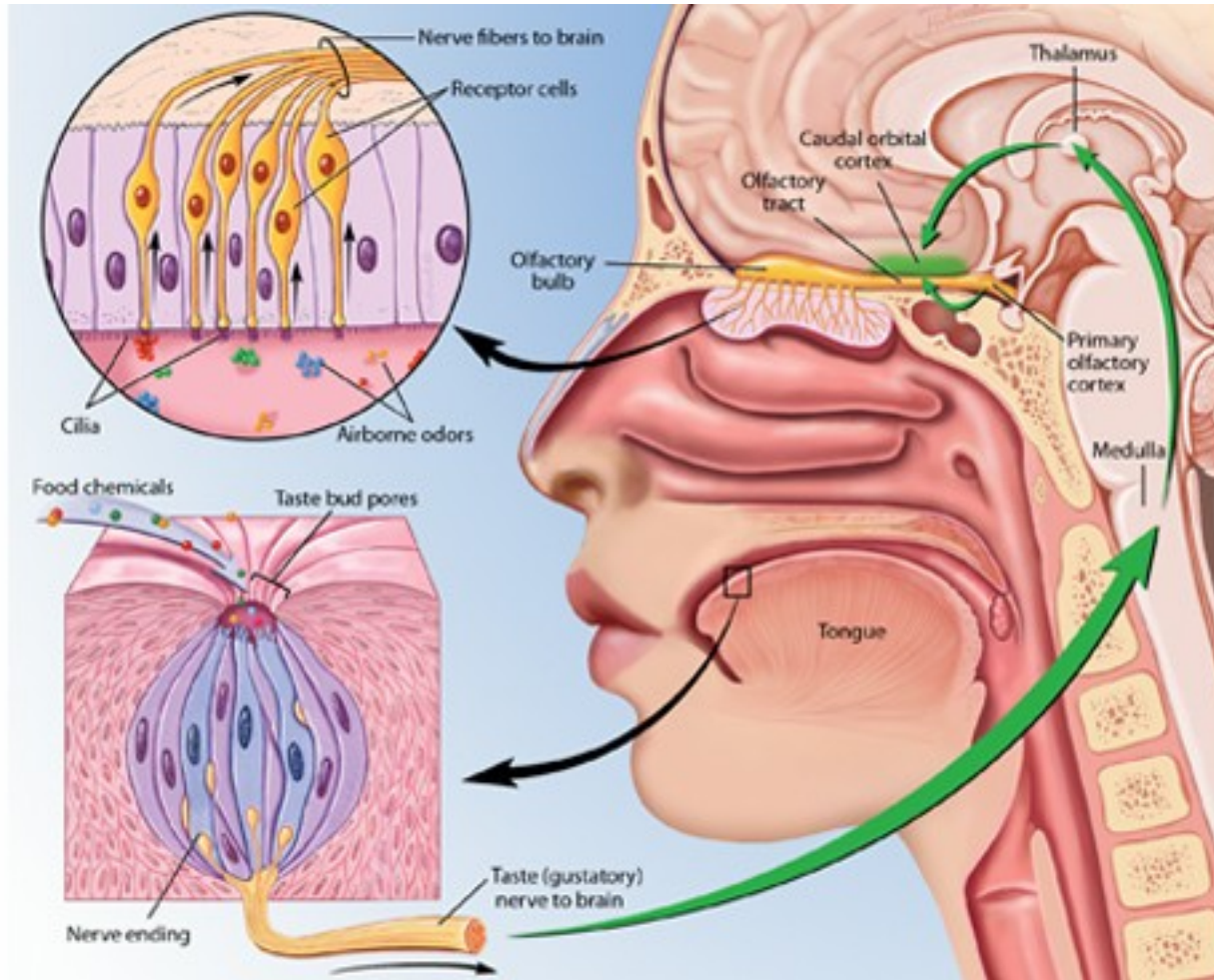
Pattern Recognition in the Brain



Chemosensory Pattern Formation

- Olfactory bulb and glomeruli
- Topographic organization
- Distributed representations

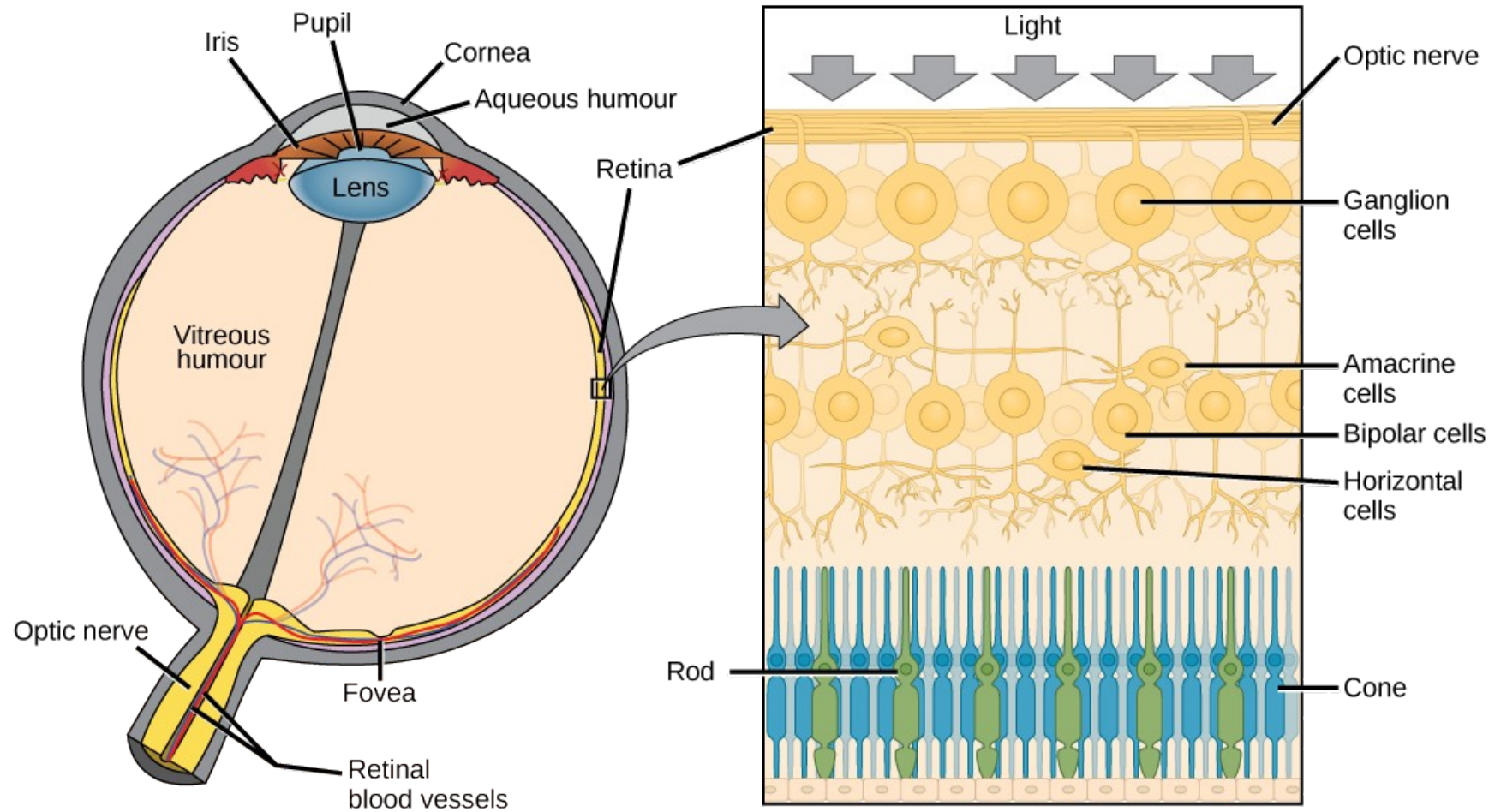
Pattern Recognition in the Brain



Chemosensory Brain Pathways

- From olfactory bulb to pallium
- Taste and internal state
- Precursors of emotion and memory

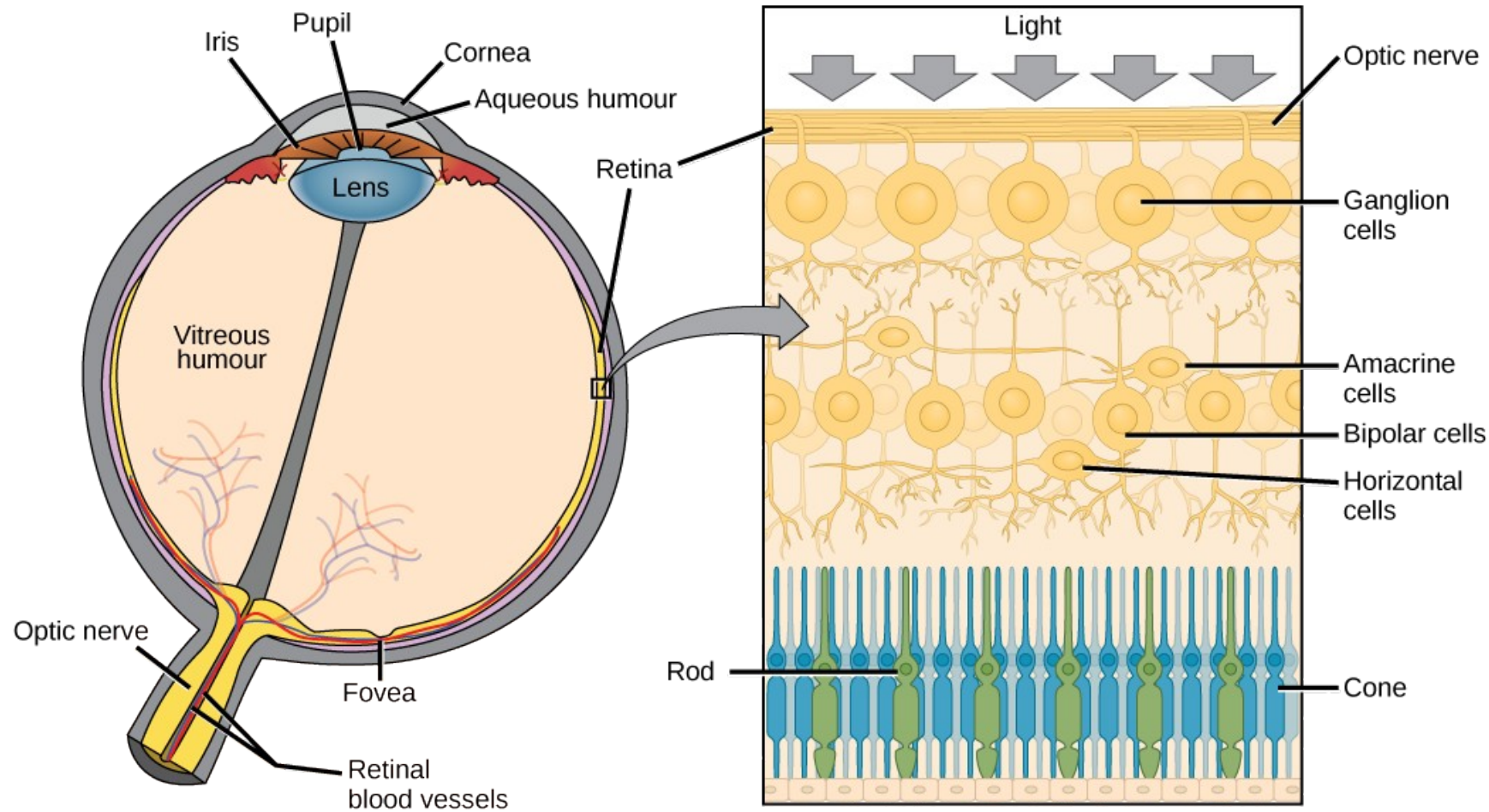
Pattern Recognition in the Brain



Retinal Transduction: Turning Light into Signals

- Photoreceptors and opsins
- Light to electricity
- Preserving spatial layout

Pattern Recognition in the Brain



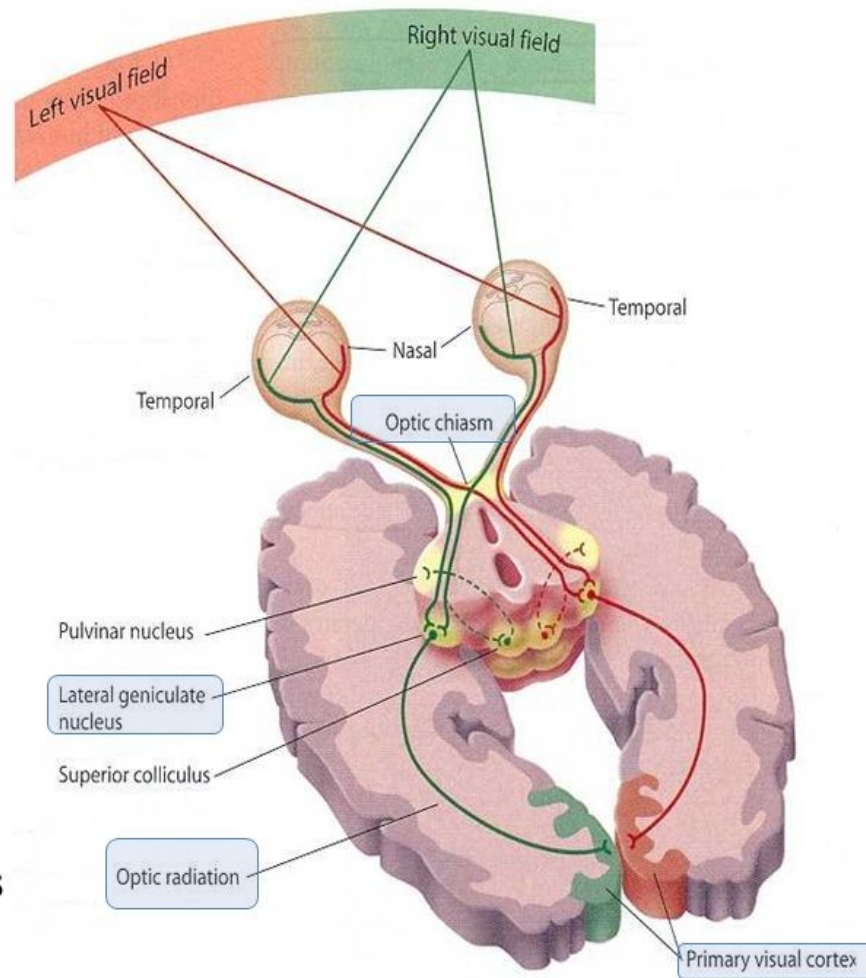
Retinal Pattern Formation

- Layered processing
- Lateral inhibition
- Edge and contrast detection

Pattern Recognition in the Brain

Visual Pathway

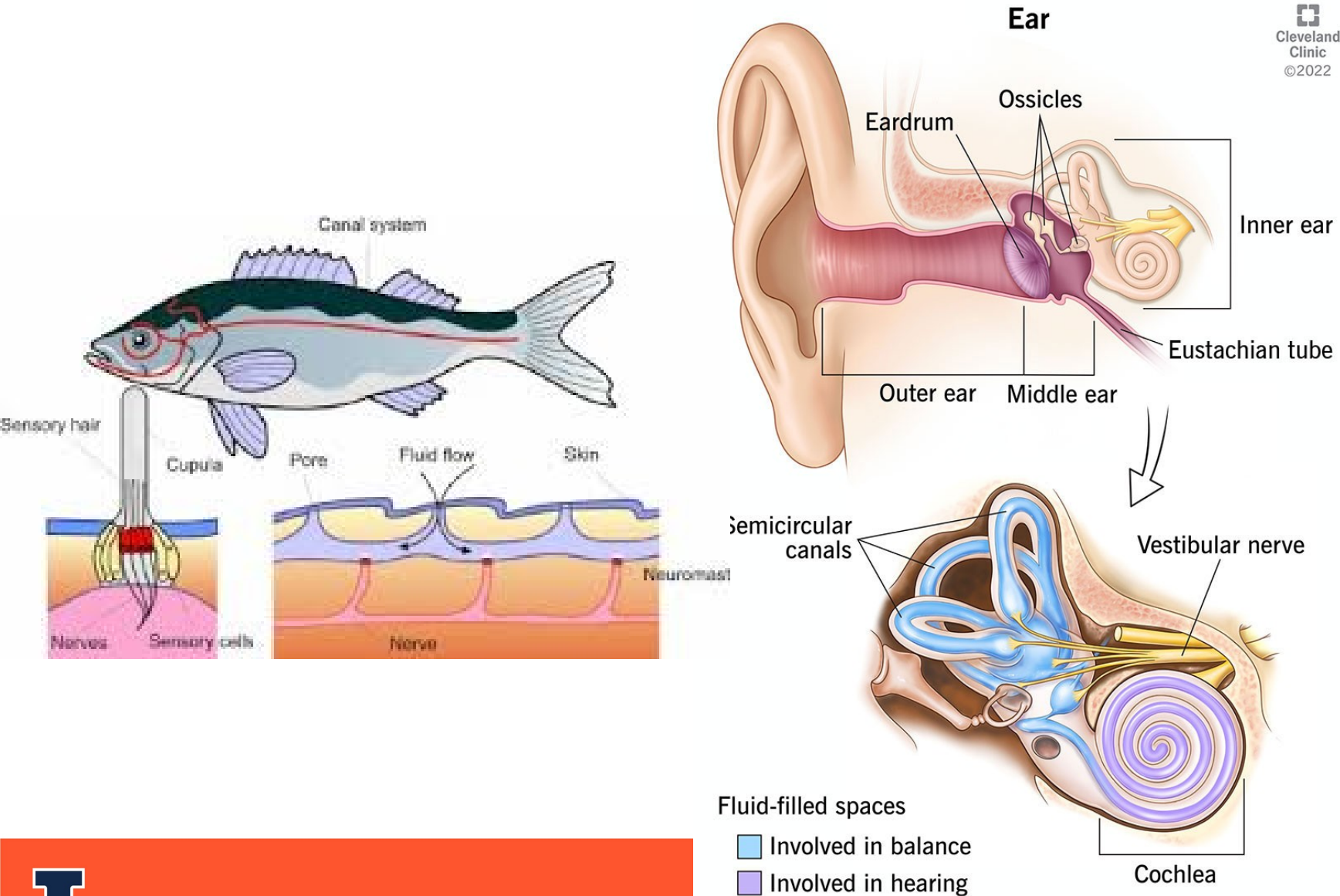
1. Cones
2. Bipolar neurons
3. Ganglion cell's axon forms the optic nerve
4. Optic nerve to the Optic Chiasm
5. Optic tract
6. Lateral geniculate nuclei of the thalamus
7. Optic Radiations
8. Primary visual areas of the occipital lobes



Early Brain Processing of Vision

- Routing through the tectum
- Visual maps and orientation
- Precursors of cortical vision

Pattern Recognition in the Brain



Auditory Transduction: From Lateral Line to Hair Cell

- Hair cells and vibration
- From water to air
- Evolutionary tinkering

Pattern Recognition in the Brain



200 Hz



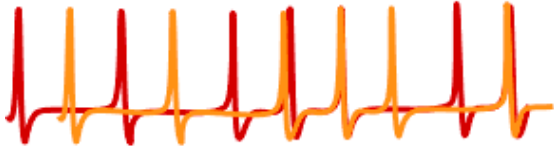
A single neuron can
encode low-frequency
vibrations



600 Hz



Individual neurons cannot
keep up with high frequencies



Phase locking enables a timing
code in the local population

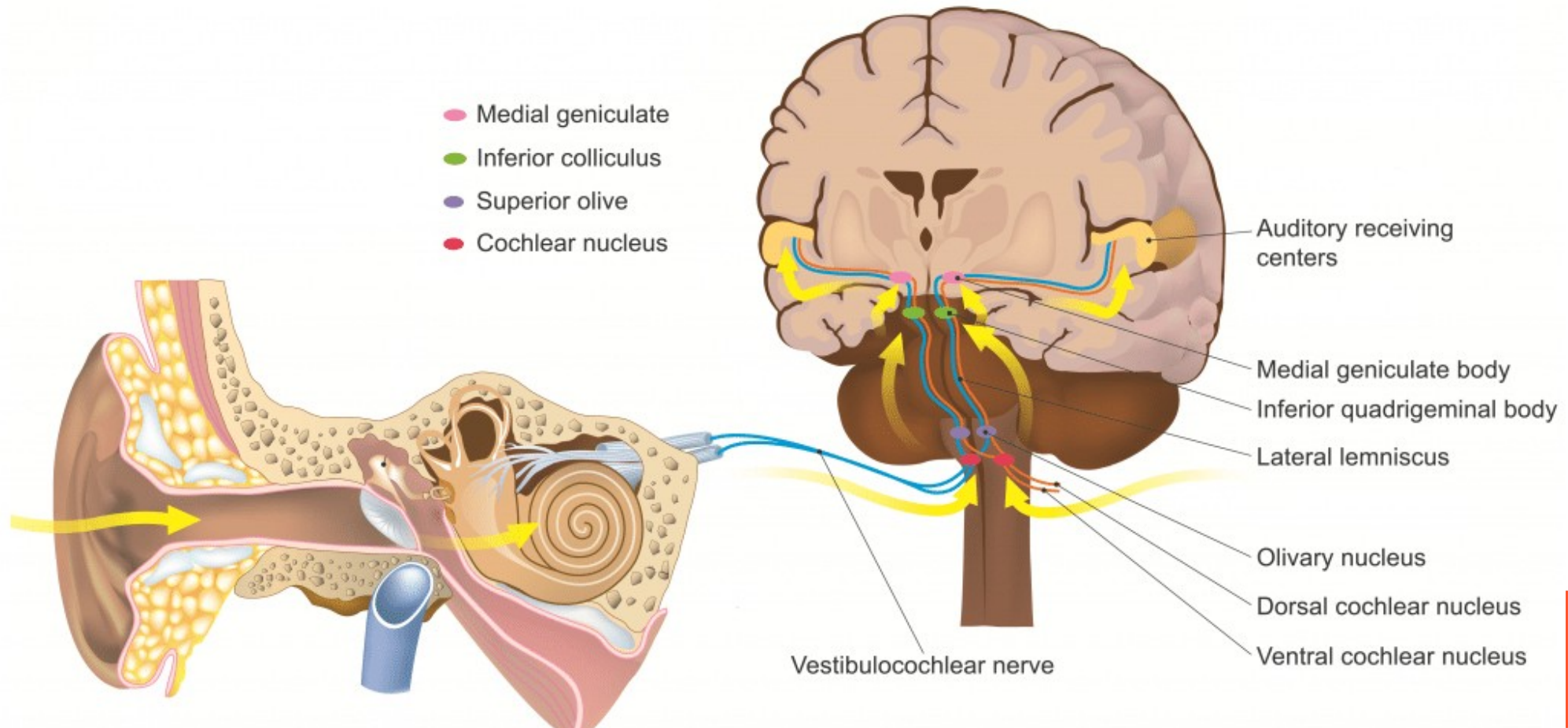
Auditory Neural Pattern Formation

- Frequency mapping
- Spatial coding of sound
- Conserved neural logic

Pattern Recognition in the Brain

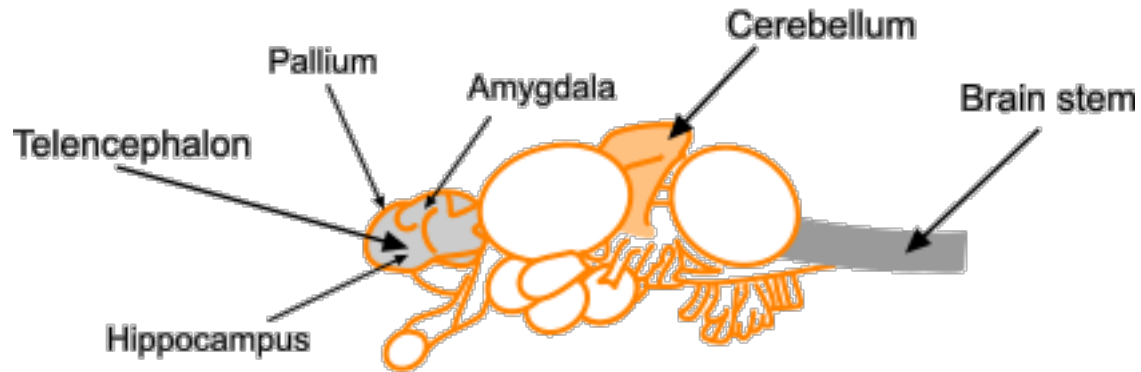
Auditory Brain Integration: Sound, Space, and Action

- Brainstem processing
- Tectal integration
- From reflex to perception



Pattern Recognition in the Brain

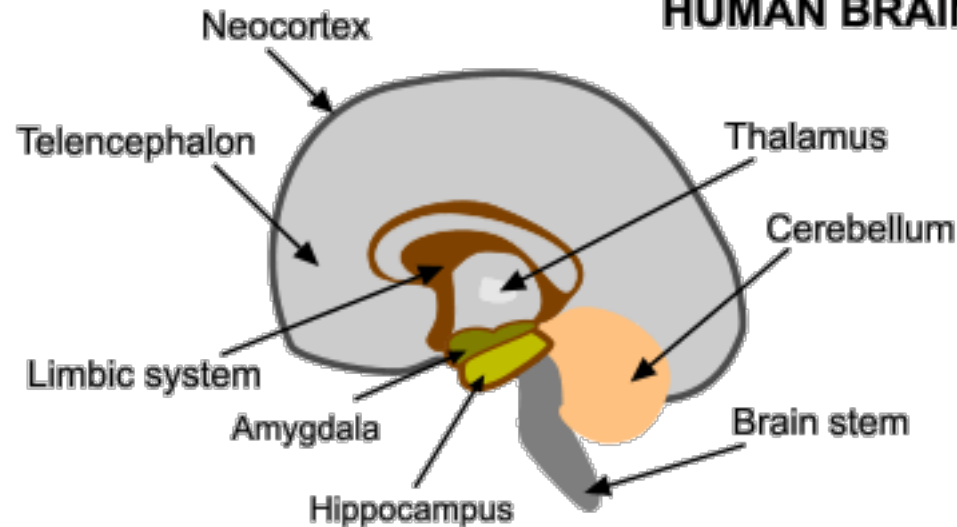
FISH BRAIN



Conserved Brain Systems for Pattern Recognition

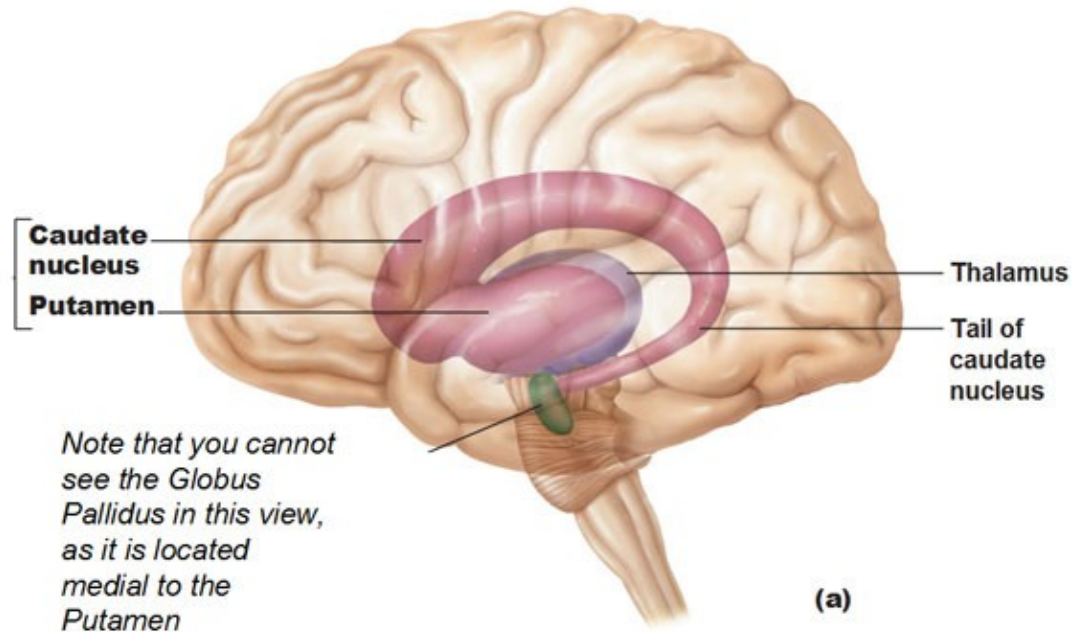
- Shared neural architecture
- Functional specialization
- Integration across systems

HUMAN BRAIN



Pattern Recognition in the Brain

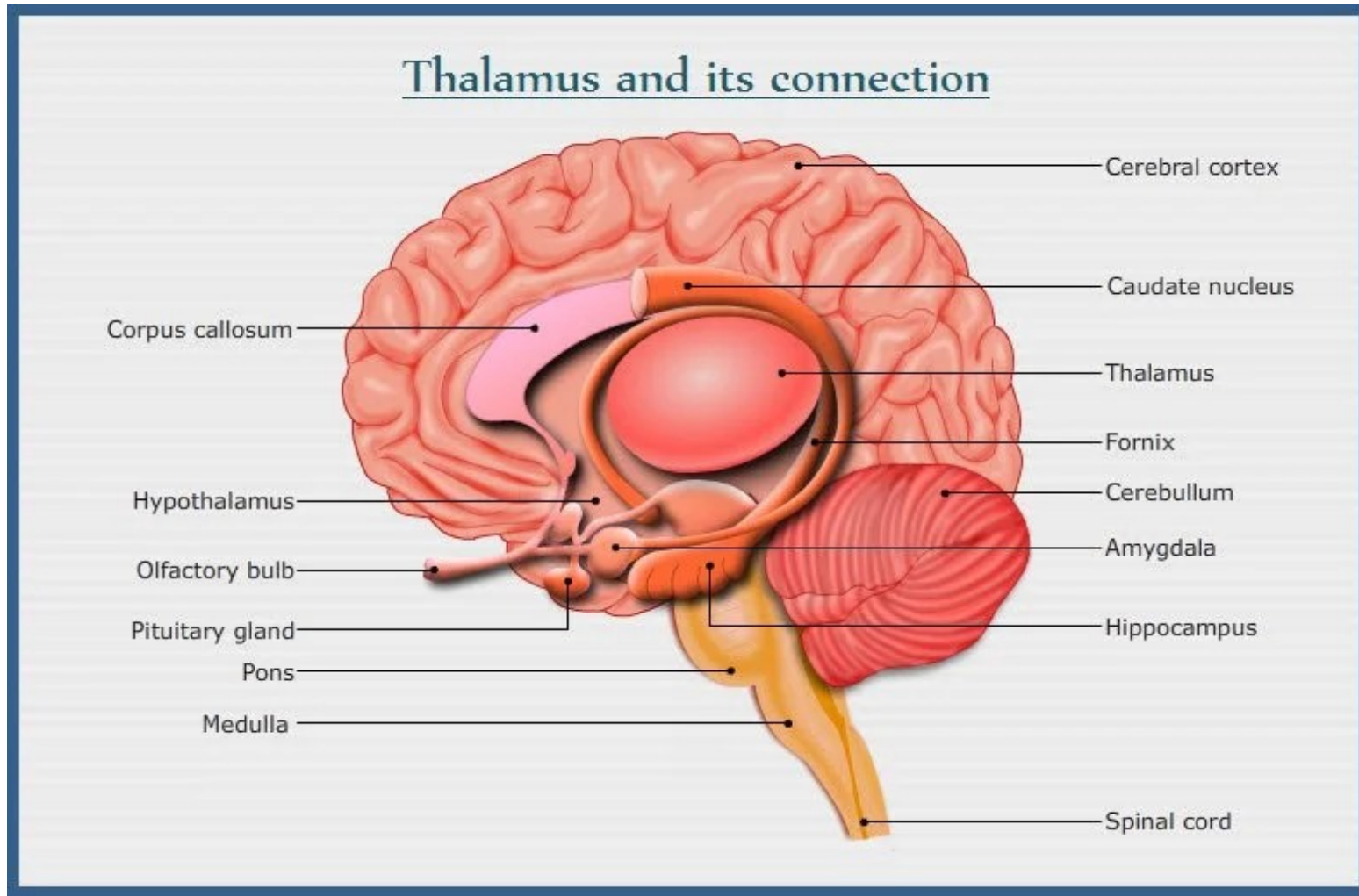
Basal Ganglia



Basal Ganglia and Context-Action Patterns

- Learning from reward
- Selecting actions
- Contextual pattern recognition

Pattern Recognition in the Brain

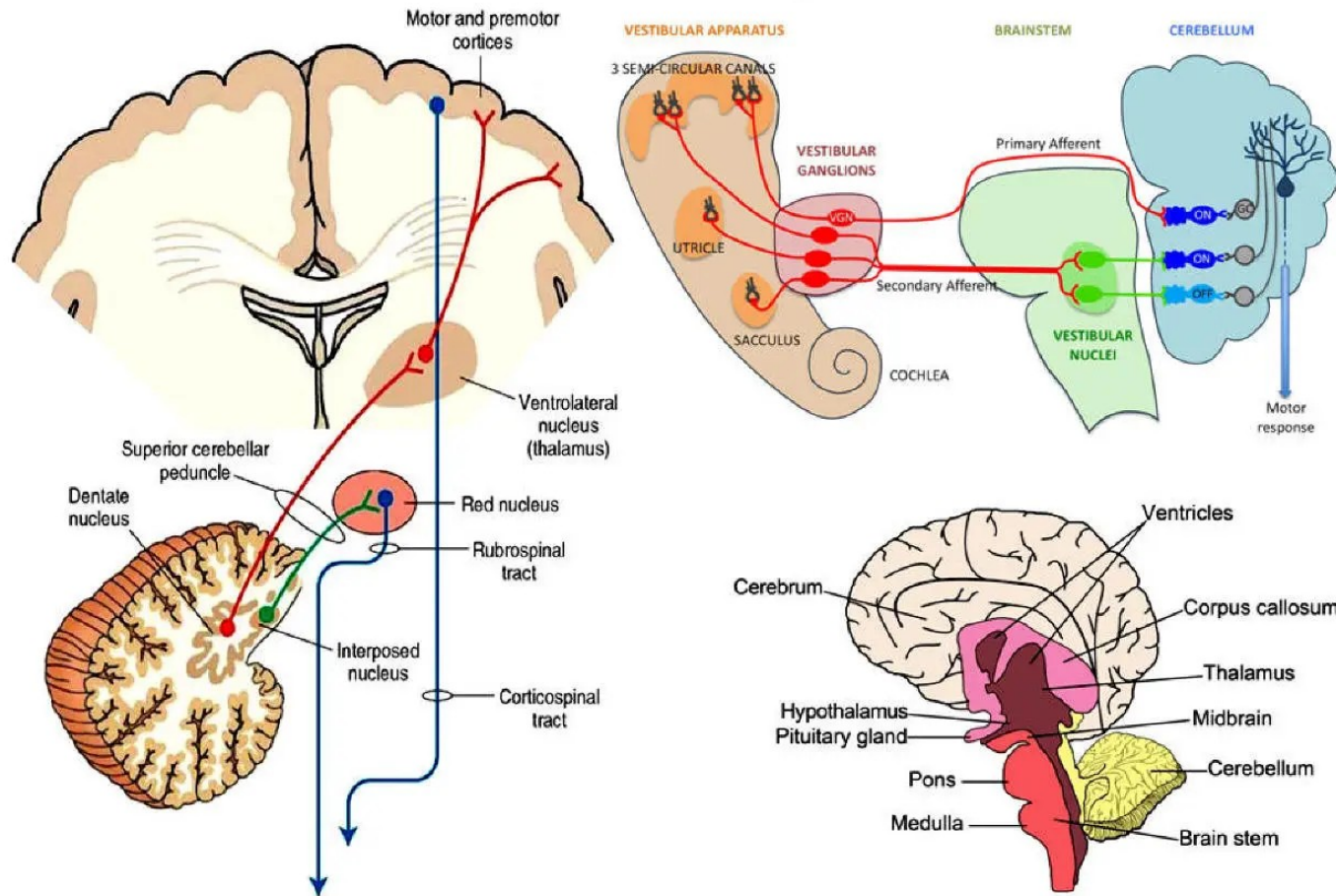


Thalamus and Sensory Gating

- Filtering and routing
- Timing and coordination
- Dynamic pattern control

Pattern Recognition in the Brain

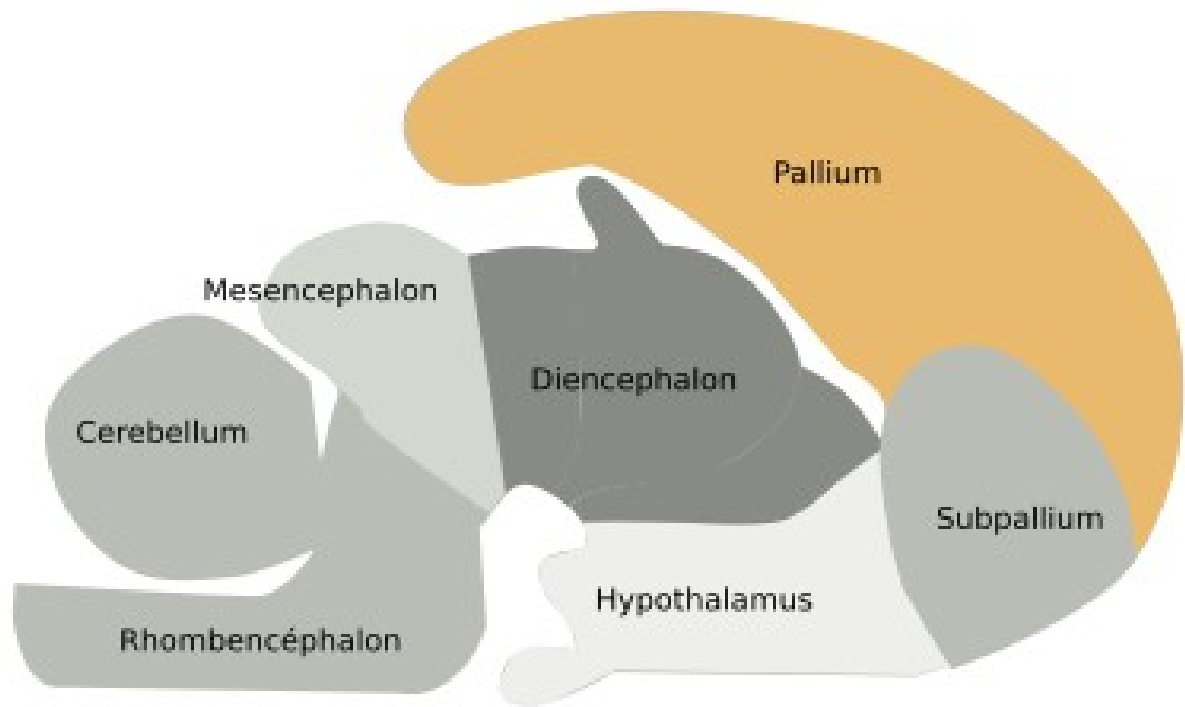
Cerebellar system



Cerebellum and Predictive Learning

- Learning from error
- Timing and coordination
- Anticipating outcomes

Pattern Recognition in the Brain

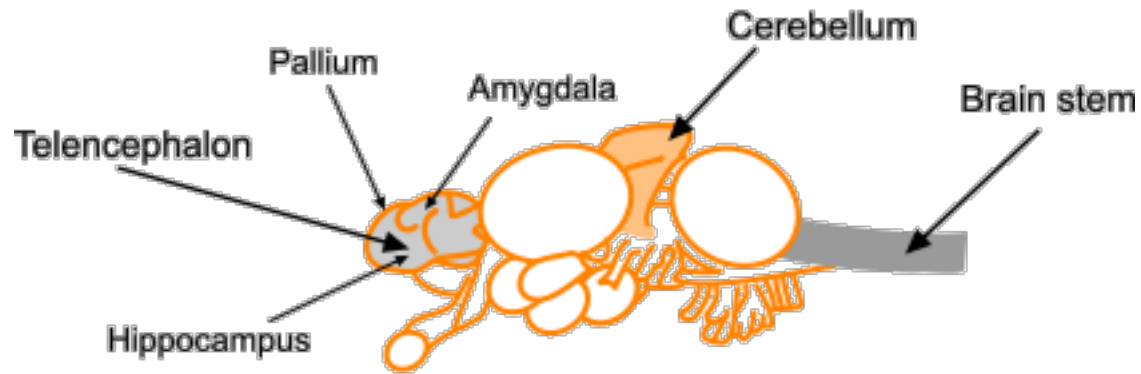


Cortex and Pallium

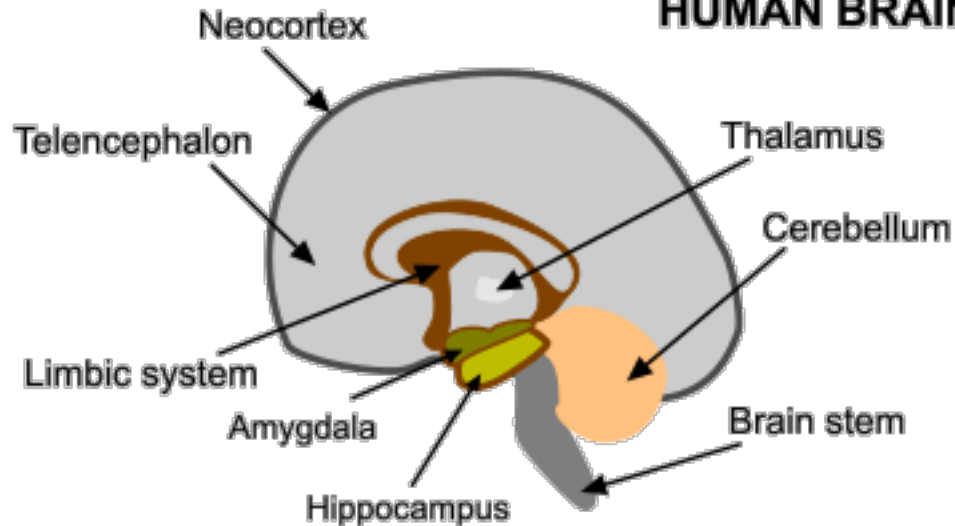
- Integrating across modalities
- Hierarchical representation
- Flexible abstraction

Pattern Recognition in the Brain

FISH BRAIN



HUMAN BRAIN



The Vertebrate Pattern-Recognition Engine

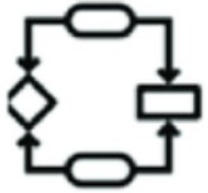
- Distributed and hierarchical
- Predictive and adaptive
- Unified design principles

Pattern Recognition in the Brain



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap

- Shared computational motifs
- Evolutionary continuity
- Brains as learning systems

Up Next

- Spatial cognition
- Curiosity, exploration, and the explore-exploit tradeoff